SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of ArtificialIntelligence and Machine Learning

ASSESSMENT1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filte segmentation techniques to ana detect edges and boundaries, an world Computer Vision problem***		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I (B VAMSI - 192472327) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: VAMSI

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula $s = \text{round}((L - 1) \times CDF)$, where $L = 256$. Determine the new intensity values and compare the contrast distribution before and after equalization.

Q2. A 3×3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3×3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Q1. Histogram, CDF, and Histogram Equalization

Pixel intensities: 10, 10, 20, 20, 20, 30, 30, 40

Total pixels $N = 8$

Histogram (frequency of each intensity):

Intensity r	Frequency n_k
10	2
20	3
30	2
40	1

Probability: $p(r_k) = n_k/N$

- $p(10) = 2/8 = 0.25$
- $p(20) = 3/8 = 0.375$
- $p(30) = 2/8 = 0.25$
- $p(40) = 1/8 = 0.125$

Cumulative Distribution Function (CDF):

- $\mathrm{CDF}(10) = 0.25$
- $\mathrm{CDF}(20) = 0.25 + 0.375 = 0.625$
- $\mathrm{CDF}(30) = 0.625 + 0.25 = 0.875$
- $\mathrm{CDF}(40) = 0.875 + 0.125 = 1.0$

Histogram Equalization ($L = 256$):

$$s_k = \mathrm{round}\big((L-1) \times \mathrm{CDF}(r_k)\big) = \mathrm{round}\big(255 \times \mathrm{CDF}(r_k)\big)$$

- $s(10) = \mathrm{round}(255 \times 0.25) = \mathrm{round}(63.75) = 64$
- $s(20) = \mathrm{round}(255 \times 0.625) = \mathrm{round}(159.375) = 159$
- $s(30) = \mathrm{round}(255 \times 0.875) = \mathrm{round}(223.125) = 223$
- $s(40) = \mathrm{round}(255 \times 1.0) = 255$

New intensity values: 64, 64, 159, 159, 159, 223, 223, 255

Contrast comparison:

Before equalization the intensities occupy a narrow range (10–40). After equalization they span almost the full dynamic range (64–255).

Q2. Averaging and Median Filters (3×3 image)

Image:

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 100 & 20 \\ 10 & 20 & 30 \end{bmatrix}$$

Neighbourhood of centre pixel (all 9 values):

10, 20, 30, 20, 100, 20, 10, 20, 30

Averaging filter (mean):

$$\frac{10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30}{9} = \frac{260}{9} \approx 28.89 \approx 29$$

(using ordinary rounding).

New centre pixel = 29.

Other pixels remain unchanged.

Median filter:

Sorted neighbourhood: 10, 10, 20, 20, 20, 20, 30, 30, 100

Median (5th value) = 20.

New centre pixel = 20.

Comparison:

- Averaging reduces the outlier (100) but also blurs the true intensity structure (centre becomes ~29).
- Median completely rejects the outlier and returns a value consistent with the surrounding region (20).

Conclusion: The median filter better preserves image details in the presence of impulse (salt-and-pepper) noise because it does not average the noise into the result.

Q3. Laplacian Filter and Sharpening

Image:

$$\begin{bmatrix} 50 & 50 & 50 \\ 50 & 100 & 50 \\ 50 & 50 & 50 \end{bmatrix}$$

Laplacian mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Response at centre pixel:

$$\begin{aligned} &0 \cdot 50 + (-1) \cdot 50 + 0 \cdot 50 \\ &+(-1) \cdot 50 + 4 \cdot 100 + (-1) \cdot 50 \\ &+0 \cdot 50 + (-1) \cdot 50 + 0 \cdot 50 \\ &= -50 - 50 + 400 - 50 - 50 = 200 \end{aligned}$$

Sharpened centre pixel (add Laplacian response to original):

$$100 + 200 = 300$$

(If the result is clipped to the 0–255 range it becomes 255; otherwise the theoretical value is 300.)

Explanation:

The Laplacian is a second-derivative operator. In uniform regions the response is near zero. At intensity transitions it produces a large positive or negative peak. Adding this peak to the original image exaggerates the transition, thereby enhancing edges and fine details (sharpening).

Q4. Global Thresholding

Image:

$$\begin{bmatrix} 40 & 60 & 120 \\ 80 & 150 & 200 \\ 30 & 100 & 180 \end{bmatrix}$$

Threshold $T = 100$(pixel $\geq T \rightarrow 1$, else 0):

Binary image:

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Foreground pixels (1s): 5

Background pixels (0s): 4

How the threshold works:

Pixels whose intensity is at least 100 are classified as object (foreground); the rest are classified as background. In this matrix the brighter regions (120, 150, 200, 100, 180) become the object.

Limitation:

A single global threshold assumes a bimodal histogram and uniform illumination. When illumination varies across the image or object and background intensities overlap, a fixed threshold produces incorrect segmentation (parts of the object may be classified as background or vice versa).

Q5. Sobel Operator – Gradient at Centre Pixel

Image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 90 \\ 10 & 50 & 90 \end{bmatrix}$$

Sobel masks:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix},\ G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Centre-pixel neighbourhood (row-major order):

$$z_1 = 10,\ z_2 = 10,\ z_3 = 10,\ z_4 = 10,\ z_5 = 50,\ z_6 = 90,\ z_7 = 10,\ z_8 = 50,\ z_9 = 90$$

Gradient components:

$$\begin{aligned} G_x &= (-1)(10) + 0 \cdot 10 + 1 \cdot 10 \\ &\quad + (-2)(10) + 0 \cdot 50 + 2 \cdot 90 \\ &\quad + (-1)(10) + 0 \cdot 50 + 1 \cdot 90 \\ &= -10 + 10 - 20 + 180 - 10 + 90 = 240 \end{aligned}$$

$$\begin{aligned} G_y &= (-1)(10) + (-2)(10) + (-1)(10) \\ &\quad + 0 \cdot 10 + 0 \cdot 50 + 0 \cdot 90 \\ &\quad + 1 \cdot 10 + 2 \cdot 50 + 1 \cdot 90 \\ &= -10 - 20 - 10 + 10 + 100 + 90 = 160 \end{aligned}$$

Gradient magnitude:

$$G = \sqrt{G_x^2 + G_y^2} = \sqrt{240^2 + 160^2} = \sqrt{57600 + 25600} = \sqrt{83200} \approx 288.4$$

Dominant edge direction:

$$\theta = \tan^{-1}\left(G_y / G_x\right) = \tan^{-1}(160/240) = \tan^{-1}(2/3) \approx 33.7^\circ$$

(relative to the horizontal axis). The edge is predominantly horizontal-to-diagonal because $G_x$is larger than G_y.

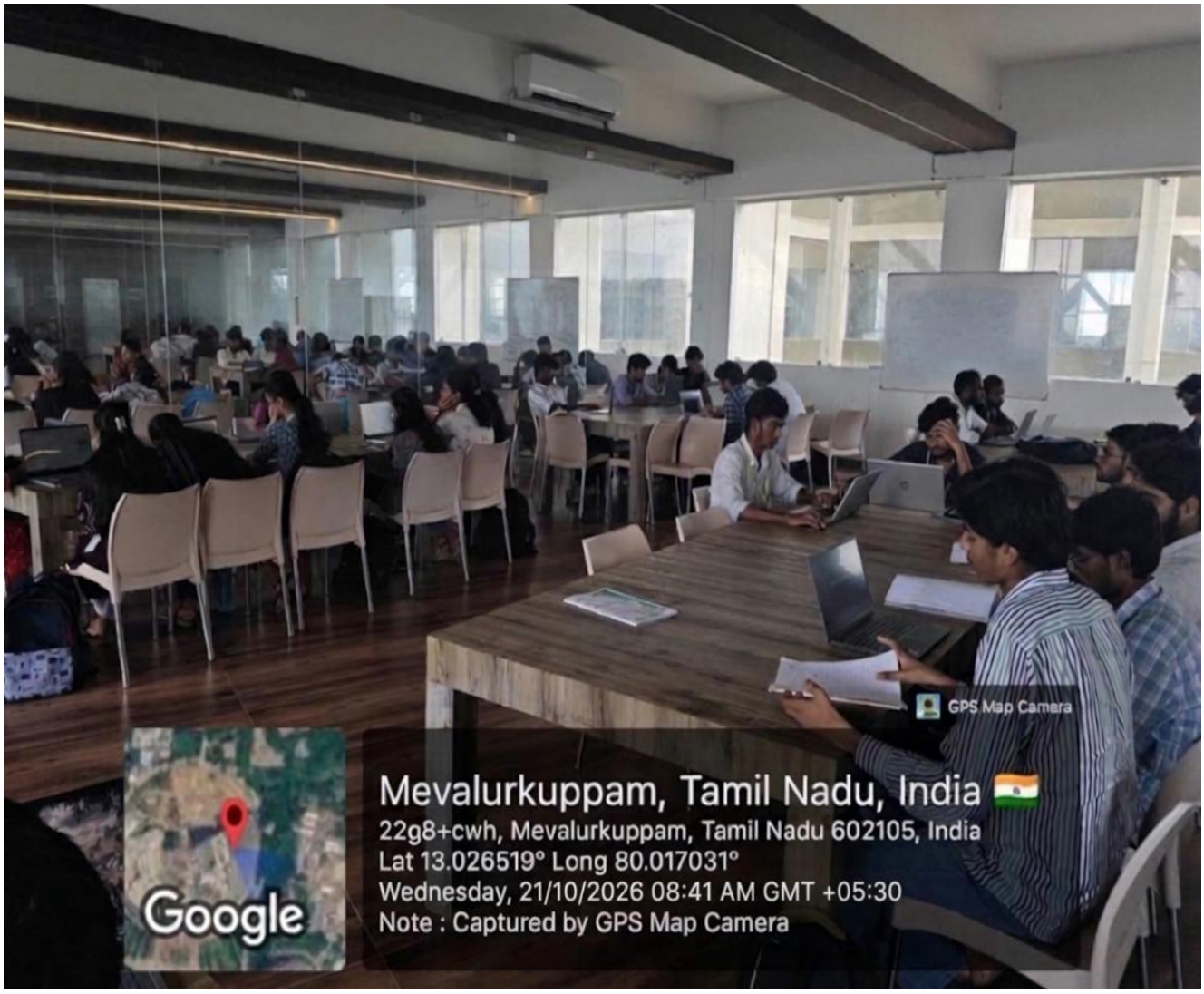
Use of edge information:

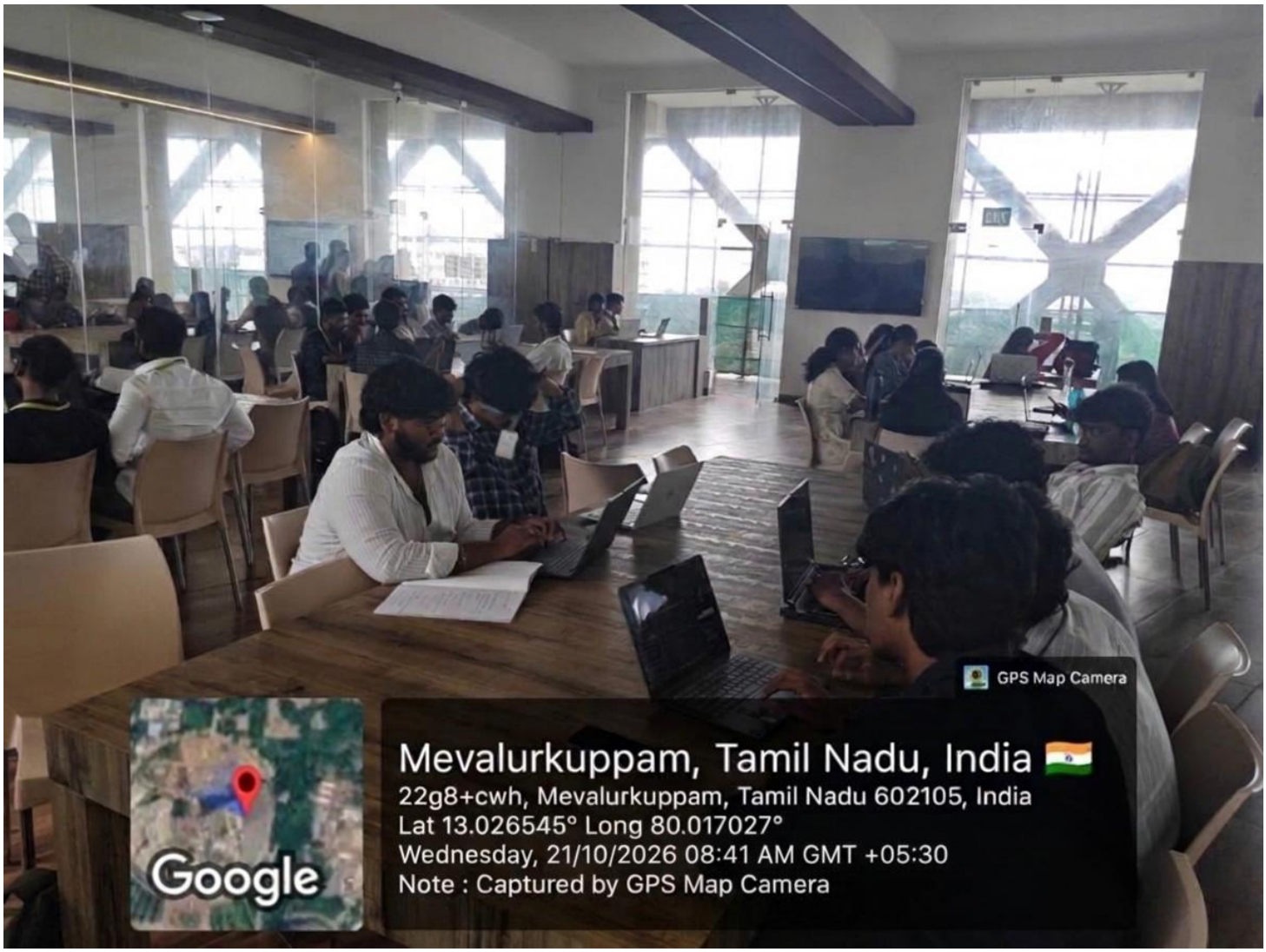
High-magnitude gradient pixels mark intensity discontinuities (object boundaries). These edge maps are the basis of many segmentation algorithms (Canny, active contours, region growing seeded from edges, etc.). The direction information can further guide edge linking and contour extraction.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator Program Director
Signature: Date:	Signature: Signature: Date: Date:




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